Mathematics: Analysis and Approaches HL

Internal Assessment

How Mathematics Determines the Volume and Surface Area of a Plastic Bottle

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1 Introduction

A lot of containers, like bottles, look simple, but are often complicated in real life, depending on their shapes. Those made with a simple straight structure, such as cylindrical or cubic, or with simple features that are easy to measure, but some of the other objects have curves. The curves on those containers make them easy for people to hold, keep, or move around, but when it comes to determining their capacity and the materials used to make them. It becomes difficult. So this means during production, they must be precise with their measurements; if not, those bottles would have different values because those mistakes don't fit their packaging standards. A small mistake in getting the Surface Area may also lead to overuse of materials, and we know how plastics are very expensive. But with calculus, you can prevent all of those small mistakes. In this IA, I will investigate how to find the volume and total surface area of a plastic bottle using mathematical modeling. I used a detergent bottle from my school dayroom, I took a picture of it, and imported it into GeoGebra. I put discrete points on one boundary of the table, then used polynomial regressions to get the equation of the curve that makes up one boundary. So, those equations that I got, I used them to calculate the volume and surface area of solids by revolution. In the end, using water and aluminium foil, I measured the capacity of the bottle, also for comparison. This was used to see how accurate the model was and to see how errors I might have made affected the results. I am interested in this topic because I was curious about calculating the measurement for irregular shapes that we use in real life, especially how industries do this. I used calculus for measurements of this bottle, which helped in understanding the link between mathematics and real-world arts and design (Dawkins, 2023). The bottle is considered symmetric at its central axis, which proves that its shape can be generated by rotating a 2D curve around the x-axis (Jerison, 2026). This IA allowed me to apply what I have learned in mathematics, which includes functions, derivatives, and integrals, as well as modeling, which I found interesting in a real object. I wanted to see how my mathematical model would compare with the actual measurements, so I used a bottle from our dayroom.

2 Aim

This exploration aims to determine the volume and total surface area of a plastic bottle using piecewise polynomial regression, integration, and solids of revolution, and to evaluate the accuracy of this model by comparing it with physical measurements. To do this, the following was done:

- construction of the mathematical model of the bottle's boundary using GeoGebra
- Integration of the model to determine the volume using the following formula:

$$V = \pi \int_a^b y^2 \, dx$$

  where: y is an equation of the bottle's boundary, for $a \leq x \leq b$ and is its interval

- Integration of the model to determine the surface area:

$$A = \int_a^b 2\pi y \, ds$$

  where $y = f(x) > 0, a \leq x \leq b$ and $ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$

- Application of the scaling factor to get the real values in centimeter units
- Verification of results using water and aluminium foil
- Where: measurement of water that can fill the bottle for volume and foil wrapping for surface area.
- Evaluation for accuracy and discussion of uncertainties.

3 MODELLING THE BOTTLE IN GeoGebra

I started by photographing the detergent bottle, ensuring that the camera was at eye level and also directly perpendicular to the bottle to ensure that I get the best image, reducing distortion. The image is shown below:

Figure 1: Photograph of the bottle

The next step was to import it into GeoGebra. I decided to add discrete points onto the graph, which will be the target points for modeling the upper curve. The polynomial regression was used because it is continuous, the same as the boundary of the bottle, and the polynomials are the best while modeling the curves, allowing us to differentiate and integrate them to calculate the volume as well as surface area (Diez et al., 2022). I tried to place the points along the boundary to get the change in the curvature, especially at the neck and at the shoulder. Those points are shown below:

Points	x	y
C	0.57	1.8
D	0.76	1.89
E	1	2
F	2	2
G	3.5	2
H	4.46	2
I	5.18	1.73
J	5.67	1.41
K	6.14	1.15
L	6.56	1.04
M	7.03	1.13
N	7.38	1.22
O	7.84	1.41
P	8.59	1.09
Q	9.17	0.79
R	9.75	0.79

Table 1: Value for 18 points for modeling the upper curve of the bottle.

4 DEVELOPING POLYNOMIAL FUNCTIONS FOR THE BOUNDARY

Due to the nature of the bottle with different curves, I managed to group the points that are consecutive into different intervals where they behave differently to get curves that are the best fit, while also ensuring that every curve is continuous with the other as [C, D, E], [E, F, G, H], [H, I, J, K], [K, L, M], [M, N, O], [O, P, Q], and [Q, R]. Using regression in GeoGebra, I found the polynomials for each interval. The highest polynomial that I got was a cubic polynomial; others are lower, which indicates that they fit well without unnecessary complications. So, the following functions were created:

$$f(x) = \begin{cases} b(x) = -0.15x^2 + 0.71x + 1.45 & 0.57 \leq x \leq 1 \\ a(x) = 2 & 1 \leq x \leq 4.46 \\ k(x) = 0.17x^3 - 2.88x^2 + 15.17x - 23.9 & 4.46 \leq x \leq 6.14 \\ l(x) = 0.49x^2 - 6.53x + 22.65 & 6.14 \leq x \leq 7.03 \\ m(x) = 0.18x^2 - 2.4x + 8.85 & 7.03 \leq x \leq 7.84 \\ n(x) = -0.08x^2 + 0.83x - 0.41 & 7.84 \leq x \leq 9.17 \\ o(x) = 0.77 & 9.17 \leq x \leq 9.75 \end{cases}$$

Those polynomials selected allow us to get the best fit of the data, ensuring the accuracy. But they are not too perfect; they may result in more oscillations. The R^2 values of those equations range between 0.901 and 0.96, which is better but not perfect, indicating there are deviations from the real shape, too. By combining all of those functions, I got the boundary of the bottle on one side, as follows:

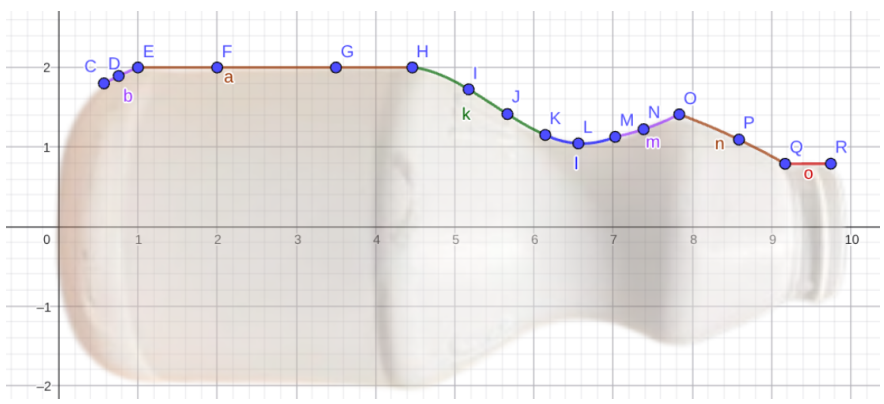

Figure 2: 18 Points (C-R) representing the Upper Curve of the bottle.

5 FINDING THE MEASUREMENTS OF THE BOTTLE

The bottle can be modeled as a solid of revolution by rotating the boundary we obtained around the x-axis to give the model of our bottle (Jerison, 2026).

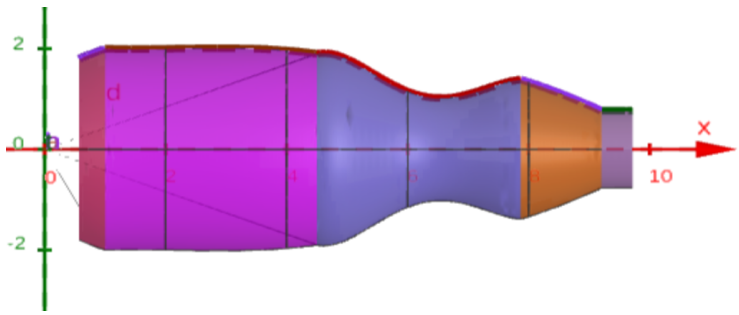

Figure 3: The model of the bottle

5.1 The volume of the bottle

To find the volume of the bottle, I will integrate the functions over the different intervals (Strang & Herman, 2016). Find the volume of a solid revolution in the interval [a,b]

$$V = \pi \int_a^b y^2 \, dx$$

Volume of a bottle =

$$V = \pi \int_{0.57}^{1} (b(x))^2\,dx + \pi \int_{1}^{4.46} (a(x))^2\,dx + \pi \int_{4.46}^{6.14} (k(x))^2\,dx + \pi \int_{6.14}^{7.03} (l(x))^2\,dx$$
$$+ \pi \int_{7.03}^{7.84} (m(x))^2\,dx + \pi \int_{7.84}^{9.17} (n(x))^2\,dx + \pi \int_{9.17}^{9.75} (o(x))^2\,dx$$

$$V = \pi \left[\int_{0.57}^{1} (-0.15x^2 + 0.71x + 1.45)^2\,dx + \int_{1}^{4.46} (2)^2\,dx \right.$$
$$+ \int_{4.46}^{6.14} (0.17x^3 - 2.88x^2 + 15.17x - 23.9)^2\,dx$$
$$+ \int_{6.14}^{7.03} (0.49x^2 - 6.53x + 22.65)^2\,dx$$
$$+ \int_{7.03}^{7.84} (0.18x^2 - 2.4x + 8.85)^2\,dx$$
$$+ \int_{7.84}^{9.17} (-0.08x^2 + 0.83x - 0.41)^2\,dx$$
$$\left. + \int_{9.17}^{9.75} (0.77)^2\,dx \right]$$

$= 4.95 + 43.48 + 5.14 + 2.42 + 2.39 + 3.20 + 1.08 = 62.65$ cubic units (2 decimal places). We have the bottle's height in the GeoGebra image as 9.75 units, while the actual height is 21cm. Because they are not equal, we will need the scale factor (Dawkins, 2023).

Scale factor $= \frac{\text{Real height}}{\text{Model height}} = \frac{21}{9.75} = 2.15$.

Then, the factor for volume will be $(2.15)^3 = 9.94$.

Thus, the real volume is $= Volume_{model} \times factor = 62.65 \times 9.94 = 622.74\,\text{cm}^3$.

5.2 The Surface Area of the Table

To get the surface area of the surface by revolution. Let's take an example of the graph of $y = f(x)$ that can be revolved around the x-axis, as shown on the next page:

Let's focus on only one slice of this solid, which is shown on the next page.

By joining the endpoints of those two circles, we get a frustrum, and we know the surface area of the frustrum as: $S = 2\pi rL$ (Wolfram.com, 2026). Where: S is surface area, r is the

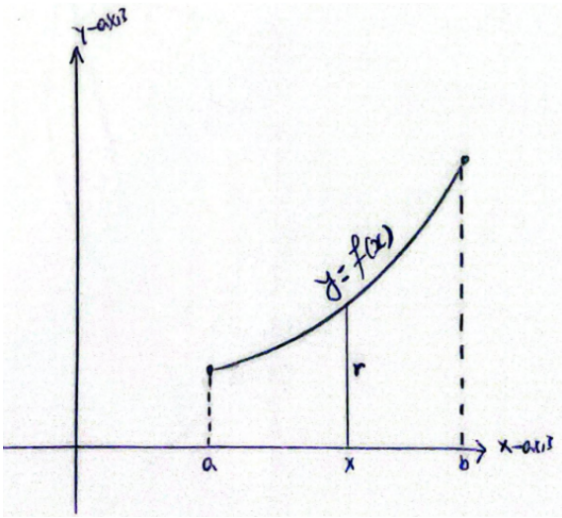

Figure 4: The curve $y = f(x)$

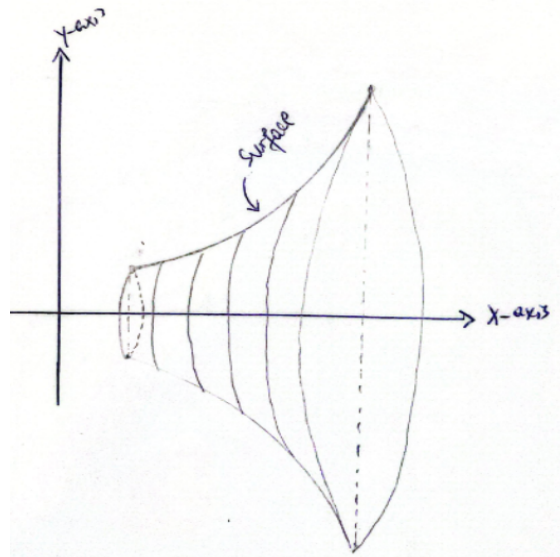

Figure 5: Solid formed by rotation of the curve $y = f(x)$ around the x-axis

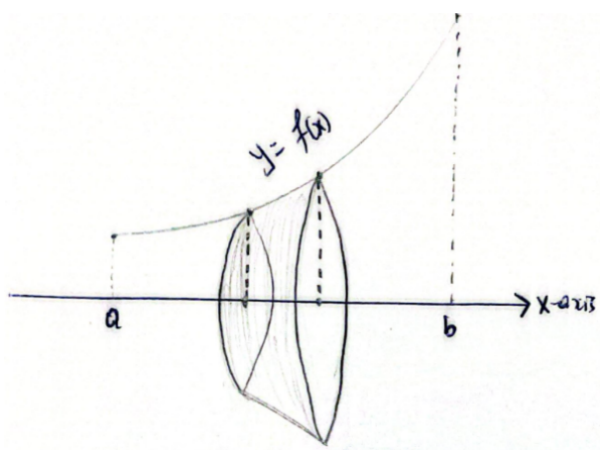

Figure 6: Frustum slice from the solid of revolution

average of those two radii, $r = \frac{r_1+r_2}{2}$, and L is the slant height.

But we know the L is not straight on the curve. Its real value can be found using the Pythagorean theorem as it is (Wolfram.com, 2026). So, $L = \sqrt{(dx)^2 + (dy)^2}$

We also know that the circumference and r can be measured at any point x in our range [a,b], which means they follow the function of our curve, f(x).

By substituting all of these in our surface area formula, we get:

$$S = 2\pi f(x)\sqrt{(dx)^2 + (dy)^2}$$

$$S = 2\pi f(x)\sqrt{1+\left(\frac{dy}{dx}\right)^2}dx$$

So the total surface area will be:

$$Area = 2\pi \int_a^b f(x)\sqrt{1+\left(\frac{dy}{dx}\right)^2}dx$$

5.3 Calculation of the area of the solid formed by our curve

We need to find the surface area, A, of a solid of revolution in the interval [a,b]: $A = \int_a^b 2\pi y\, ds$ where $y = f(x) > 0, a \leq x \leq b$ and $ds = \sqrt{1+\left(\frac{dy}{dx}\right)^2}dx$. This formula is the sum of all the curved surface area obtained when the function was rotated, while the $\sqrt{1+\left(\frac{dy}{dx}\right)^2}$ gives the slope of the curve to make sure that the surface area will follow the real shape of the bottle (Dawkins, 2023). The derivatives of our functions are:

- $b'(x) = -0.30x + 0.71$
- $a'(x) = 0$
- $k'(x) = 0.51x^2 - 5.76x + 15.17$
- $l'(x) = 0.98x - 6.53$
- $m'(x) = 0.36x - 2.4$
- $n'(x) = -0.16x + 0.83$
- $o'(x) = 0$

The total surface area of a bottle $= A_b + A_a + A_k + A_l + A_m + A_n + A_o$

$$= \int_{0.57}^{1} 2\pi(-0.15x^2 + 0.71x + 1.45)\sqrt{1+(-0.30x+0.71)^2}dx + \int_{1}^{4.46} 2\pi(2)\sqrt{1+0^2}dx$$

$$+ \int_{4.46}^{6.14} 2\pi(0.17x^3 - 2.88x^2 + 15.17x - 23.9)\sqrt{1+(0.51x^2 - 5.76x + 15.17)^2}dx$$

$$+\int_{6.14}^{7.03} 2\pi(0.49x^2 - 6.53x + 22.65)\sqrt{1 + (0.98x - 6.53)^2}dx$$

$$+\int_{7.03}^{7.84} 2\pi(0.18x^2 - 2.4x + 8.85)\sqrt{1 + (0.36x - 2.4)^2}dx$$

$$+\int_{7.84}^{9.17} 2\pi(-0.08x^2 + 0.83x - 0.41)\sqrt{1 + (-0.16x + 0.83)^2}dx + \int_{9.17}^{9.75} 2\pi(0.77)\sqrt{1 + 0^2}dx$$

$= 5.72 + 43.48 + 12.00 + 5.38 + 5.12 + 8.01 + 2.81 = 82.52$ square units

I can say that what we got above is the lateral surface area. But also, there is the base, which must be included in the area. The area of the base can be found using the formula: $A = \pi(b(x))^2$ where A is the Area and $b(x)$ is the radius of the circle and x is where that circle lies. So, $A = \pi(-0.15(0.57)^2 + 0.71(0.57) + 1.45)^2 = 10.25\,\text{cm}^2$. Then the total surface area is this: $10.25 + 82.52 = 92.77\,\text{cm}^2$. As we have seen, when we were calculating the volume, the scale factor was 2.15. So the factor for the Total surface area is $2.15^2 = 4.62$ Thus, the Total Surface Area $= Area_{model} \times factor = 92.77 \times 4.62 = 428.60\,\text{cm}^2$.

6 VERIFICATION OF RESULTS

So, after getting the values of Volume and Total Surface Area. I'm going to do the bottle's measurements manually. This will help me ensure that my model reflects the real physical bottle. I did this in the following two ways:

6.1 Verification of the Volume

I filled the bottle with water until it was full. As shown below:

After feeling it, I transferred the water into the measuring cylinder, then I recorded the volume, which was found to be almost 625 cm^3. This is very close to the volume I obtained from the bottle's model, 622.74 cm^3.

Difference $= 625 - 622.74 = 2.26\,\text{cm}^3$.

Percentage error $= \frac{2.26}{625} \times 100\%$ (Math Is Fun, 2025).

Figure 7: The water bottle is being filled with water to be measured.

So, this is the percentage error, and the small difference shows that the mathematical calculation is accurate.

6.2 Verification of the Total Surface Area

To get the real value of the Total Surface Area of this bottle. I wrapped the Aluminium foil around this bottle in different parts that I had divided through this bottle for the better results as follows:

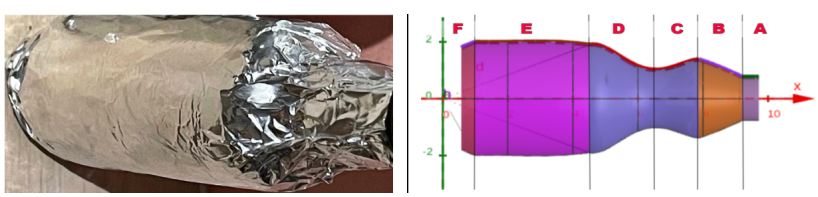

Figure 8: Image of the bottle wrapped in the Aluminium foil showing divisions.

The parts are shown in capital red colors as A, B, C, D, E, F, and also G, which is not visible in the image but represents the base, located at the bottom, or the part on which a bottle stands. After cutting it, I found that those small, regular shapes were joined together. A and E are cylinders; B, C, D, and F are frustums; and G is a cylinder that forms the bottom. So, I found the area of every part using those relevant formulas:

- Circumference, $C = 2\pi r$

- Radius from circumference, $r = \frac{C}{2\pi}$
- Cylinder lateral area, $A_{cyl} = Ch$
- Frustrum lateral Area, $A_{frus} = \pi(r_1 + r_2)s$
- Where Slant height, $s = \sqrt{(r_1 - r_2)^2 + h^2}$ and $r_i = \frac{C_i}{2\pi}$

The circumferences are the actual ones that were measured directly from the bottle. Then, this is the table of the information that I calculated, and also showing all the information that I got from the measurement I made on the bottle:

Shape	Circumference(s) (cm)	Radii(cm)	Height(cm)	Slant(cm)	Area(cm2)
A	11.50	$r = 1.83$	0.9	-	10.35
B	19.50, 11.50	$r_1 = 3.10, r_2 = 1.83$	2.1	2.46	38.07
C	15.50, 19.50	$r_1 = 2.47, r_2 = 3.10$	2.0	2.10	36.73
D	28.70, 15.50	$r_1 = 4.57, r_2 = 2.47$	3.2	3.83	84.60
E	28.70	$r = 4.57$	6.3	-	180.81
F	26.50, 28.70	$r_1 = 4.22, r_2 = 4.57$	0.7	0.78	21.60
G	26.50	$r = 4.22$	-	-	55.88
Total	-	-	-	-	428.02

The Total Surface Area is $428.02\,\text{cm}^2$. Let us check the difference and the percentage error.

Difference $= 428.60 - 428.02 = 0.58\,\text{cm}^2$.

Percentage error $= \frac{0.58}{428.02} \times 100 = 0.14\%$.

The difference and the percentage error show that the area that was calculated using the model and the area that was calculated manually are very close, where the difference should have arisen from the rounding of some values and other minor effects.

7 Errors, Uncertainties, and Limitations of this Exploration

The following are sources of the errors that were likely to have occurred during the process of calculating the volume and the surface area of the bottle. A ruler was used to measure the dimensions, such as height; it has a natural uncertainty of ± 0.1 cm. When the volume is the cube of the dimensions, the surface area is twice the dimensions. This indicates that even a small error would result in a large one when it comes to the final results of the area and the volume. Distortion effects could also apply to the image that was used in the GeoGebra, as you can not be able to be sure that the camera is in the right position, which could result in compression or stretching. Placing the points around the boundary of the bottle in GeoGebra, the points could be off the boundary by milliliters, which affected the polynomial regressions, and also, they are just approximations of the best-fit line, not the true bottle curve, even though the R^2 values were high, but they are not 100% accurate. During verification, I was measuring the water level, which had some source errors, such as reading the meniscus, bubble formation, and water sticking on the walls. The aluminium foil could also stretch or fold, being another source of error. I was rounding all of the values to two decimal places, which would have limited the accuracy too. Despite all of those limitations, the model calculations of the capacity matched the physical measurements, where they were very close together, which indicates the strength of the modeling process.

8 Evaluation of the Model

As we have seen, the model was successful in comparison to the produced values and real measurements; however, as shown above, it still exhibits some limitations. I used the piecewise polynomial functions to make sure the model provided is according to the curvature of its boundary, but also, it doesn't guarantee that every function is over the curvature of

the bottle. Another limitation was that the bottle is perfectly rotationally symmetric, as some bottles in real life may have small irregularities that can't be presented by rotating a 2D curve around the x-axis. Despite all of those weaknesses, it also showed strengths; the percentage error by comparing physical measurements and capacity from the model was less than 1% for the volume and less than 0.2% for the surface area. I used consistency throughout integrals, scaling factor, and physical measurements, indicating the approach for this IA was appropriate. So, to further improve the model, using spline or Bézier curves with more points could yield a more accurate fit, and using digital calipers could reduce measurement uncertainty. These suggestions would improve this model more precisely, but this model is also good enough to prove the power of calculus in real-life measurements.

9 Reflection

This exploration shaped my understanding of how mathematics is applied in real-life situations and the importance of modeling and critical thinking in Mathematics. I knew before that integrals are used in class as formulas, but through this IA, I saw their application in manufacturing, like calculating the capacity and area of irregular containers. I also found that mathematical models always require assumptions and simplifications, so understanding their limitations is essential when performing calculations. I also learned to maintain accuracy throughout my work, such as using a piecewise model and taking measurements. This modulation also improved my understanding of calculus and solids of revolution. This IA made my knowledge of Mathematics meaningful as it helped me apply what I learned and what I know in real-world problems, like in industries where decisions are made according to mathematics in the formation of bottles.

10 Conclusion

This exploration aimed to determine the exact volume and total surface area of a plastic detergent bottle using functions, integrals, and mathematical modeling, and to compare these results with physical measurements. After modeling the bottle's shape using piecewise polynomials and integrating them for its volume and surface area, I found a volume of 622.74 cm^3 and a surface area of 428.60 cm^2. While for the physical measurements, I found that the volume of the bottle is 625 cm^3 and the surface area of 428.02 cm^2, indicating a small difference of less than 1% for each. This proves that the model was very accurate, and it also shows that calculus is the best for analysing irregularly shaped objects. This exploration had several limitations, but the close results from physical and model measurements prove that the model was very appropriate. In short, this exploration reached its aim and also showed how mathematics can be applied in real life.

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